Inkspection: Detecting Document Forgery by Cross-Checking Pixel, Typographic and Semantic Evidence at the Word Level

Abdul Wasay and Muhammad Taha
COMSATS University Islamabad, Wah Campus
Wah Cantt, Pakistan

***Abstract*—Receipts and invoices are increasingly scanned, read by OCR and paid without close human inspection, so a small edit to a total or date can easily slip through. Pixel-forensic detectors find where an image was touched but not whether the change makes the document wrong. Rule-based checks catch totals that don't add up but miss edits that keep the arithmetic intact, and they mistake OCR misreads for fraud. We present Inkspection, a system of cooperating agents that collect pixel, typographic and semantic evidence and tie it to the same OCR words. A calibrated word-level model scores the document, marks the altered regions and writes an audit report that cites its evidence. On the Find It Again! benchmark, Inkspection reaches a document-level PR-AUC of [•] versus [•] for the strongest single-modality baseline, and a word-level localisation F1 of [•]. At 95% recall it sends [•]% of documents to human review instead of [•]%.**

***Keywords*—document forgery detection, image forensics, optical character recognition, multimodal fusion, multi-agent systems, explainable AI**

I. Introduction

Expense claims and supplier invoices now arrive as scans or phone photos, are read by OCR and are paid automatically, often without anyone looking at the image. A few minutes in an ordinary image editor is enough to raise an amount or move a date. Such edits are hard to catch because they are small: forgers change a few characters and try to make them blend in. In the Find It Again! dataset [1], built from SROIE receipts [4], volunteers forged receipts with everyday tools such as Paint and GIMP, and in 353 of 455 modified areas they simply copied characters from elsewhere on the same receipt. An edit like this leaves visual traces, such as a patch that compresses differently or a digit drawn thinner than its neighbours, and logical ones, such as items that no longer add up to the total.

Most detectors look for only one kind. Pixel-forensic methods, from error level analysis to learned tampered-text detectors [2], find where pixels changed but cannot tell whether a change matters, and they react to innocent stamps, folds and pen marks. Rule-based validators check the text produced by OCR systems such as PP-OCR [5] or TrOCR [3]. They catch sums and dates that disagree, but a careful edit that keeps the arithmetic consistent passes, and a misread digit triggers a false alarm just as an edited one does.

We argue that the two kinds of evidence should be checked against each other, word by word. An amount that breaks the arithmetic and is also printed thinner, sits off the baseline and recompresses differently from its neighbours is almost certainly forged. A failed sum on a word the OCR engine was unsure about may be nothing. Page-level pixel and text scores cannot tell these cases apart.

Inkspection splits the work between agents. The orchestrator keeps the original image untouched for forensics and makes a straightened copy for OCR. The vision agent runs six pixel-forensic detectors on the original while the OCR and layout agent reads the words and compares each word's typography with the rest of its line. The semantic agent extracts the receipt's fields and checks quantities, sums, tax, change and dates, recording which words each failed check involves. The adjudicator fuses all evidence word by word with a calibrated gradient-boosted model and returns a document score, a review tier, the suspicious regions and an audit report.

We compare Inkspection with four traditional detectors: an error level analysis threshold, a vision-only classifier, a rules-only validator and a naive fusion of the last two. All systems are scored from the same cached agent outputs, so differences come only from how evidence is combined. Besides detection and localisation metrics, we report the share of documents each system sends to human review at 95% recall.

Our contributions are:

1) A multi-agent architecture that links pixel-forensic, typographic and semantic evidence to the same OCR words, with forensics always run on the unmodified image.
2) A calibrated word-level fusion model with cross-modal agreement features that localises altered regions and explains each through its evidence.
3) An evaluation on Find It Again! against single-modality and naive-fusion baselines on identical inputs, reporting review workload at fixed recall and catch rates at an equal false-alarm rate.
4) An ablation of each evidence source, and an open-source implementation with a web interface for analysis and review.

Section II reviews related work, Section III describes the agents and fusion model, Section IV the dataset, baselines and metrics, Section V the results, and Section VI limitations and future work.

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